

Ali Nikkhah

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TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, JAX, TensorFlow, Hugging Face, Time-Series, ONNX Runtime
Agentic AI & Advanced RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, Graph RAG, KAG
Vision & Audio Systems: OpenCV, BLIP, Vision Transformers (ViT), T5, Whisper, wav2vec2.0, Librosa, ESPnet
Data Engineering & Lakes: Apache Spark, Trino, ClickHouse, Airflow, PeerDB, Apache Kafka, RabbitMQ, Parquet
MLOps & Infrastructure: Ray, Kubeflow, Triton Inference Server, TensorFlow Serving, TensorRT-LLM, TF Lite
DevOps & Automation: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, GitLab CI
Monitoring, Governance & BI: Prometheus, Grafana, ELK Stack, Metabase, Weights & Biases, MLflow, DVC

PROFESSIONAL EXPERIENCE

Turquoise Digital

Tehran, Iran

Senior Data Scientist & AI Engineer

Aug 2025 - Mar 2026

- Engineered an enterprise telephony voice intelligence pipeline, utilizing transformer-based speech emotion recognition architectures (Whisper, wav2vec2.0) to capture, transcribe, and analyze live audio data streams.
- Aligned historical time-series pricing vectors with high-frequency market sentiment news text embeddings, deploying deep sequence models to map cross-modality pricing correlations and market turbulence effects.
- Built asynchronous audio streaming workers utilizing Python's Asyncio library to process parallel incoming voice payloads without blocking main thread API endpoints.
- Optimized downstream model latency by compiling acoustic feature extraction blocks directly into TorchScript, accelerating runtime execution across automated ticketing clusters.
- Configured continuous validation frameworks to monitor acoustic signal-to-noise ratios, dynamically adjusting transcription thresholds under poor telecommunication connectivity conditions.

Digikala

Tehran, Iran

Senior AI Engineer

Jan 2025 - Apr 2025

- Led the development of multimodal retrieval systems, coordinating the deployment of vision-language architectures to index, align, and fetch non-homogenous asset modalities (images, diagrams, and text product profiles).
- Integrated Vision Transformers (ViT) and convolutional feature extractors into core search pipelines, establishing shared vector spaces to enable text-to-image semantic searching.
- Programmed high-throughput image preprocessing and spatial augmentation workers, utilizing optimized libraries to decouple raw upload streams from critical inference execution blocks.
- Conducted comprehensive performance profiling evaluations on multimodal alignment frameworks, optimizing matrix operations to prevent out-of-memory (OOM) failures under heavy traffic bursts.
- Supervised the deployment of multi-tenant model execution clusters, leveraging Kubernetes to isolate vision model computation nodes from standard language generation clusters.

SELECTED PROJECTS & RESEARCH

Advanced Analytics Australia

Remote Contract

Computer Vision Engineer

Oct 2024 - Apr 2025

- Engineered edge and server-side video analytics pipelines on AXIS camera systems to automate health, safety, and compliance auditing, including handwash tracking, protective gear detection, and restricted zone entry monitoring.
- Optimized deep vision network deployment across resource-constrained edge computing hardware by utilizing static model quantization, layer compilation, and concurrent container hosting configurations via Triton Inference Server and TensorFlow Lite.
- Architected a high-throughput spatial metadata ingestion broker using Apache Kafka combined with optimized GPU memory caching routines to handle concurrent high-frame-rate tracking streams and dynamic sensor calibration.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering - Digital Systems (GPA: 3.65, Major: 3.91)

Sept 2020 - Jul 2024